KV Quantization Drift in Flow-Matching VLAs

Project objective. We are benchmarking whether KV-cache and attention-state quantization can reduce VRAM during VLA inference without degrading LIBERO task success. Based on the revised 5/23 scope, the project is now framed around a narrower hypothesis: prefix-suffix VLAs such as pi0.5 and cross-attention diffusion/flow VLAs such as GR00T N1.7 expose different quantization failure modes, so a single static low-bit calibration is likely insufficient.

Experiments conducted so far. We built a Modal-based LIBERO evaluation pipeline for pi0.5/OpenPI and Isaac GR00T N1.7, targeting reproducible LIBERO-Spatial rollouts, result logs, videos, and GPU memory measurements. The pi0.5 smoke experiment successfully completed LIBERO-Spatial with 1 rollout per task: 10 episodes total, 10 successes, 100% success rate. This verifies that the simulator, model server, result volume, and video-output path work end-to-end. We also generated a quantization-drift probe figure that reports per-denoising-step RMS drift, relative drift from step 0, and static-versus-per-step quantization MSE. The current figure uses synthetic tensors matching the expected failure pattern and will be replaced with logged pi0.5/GR00T tensors once hooks are attached.

GR00T N1.7 has been attempted but is not yet fully benchmarked. The setup exposed several practical blockers: LIBERO submodule assumptions inside Modal, uv CLI incompatibilities, missing checkpoint-file handling, and gated Hugging Face dependency access for `nvidia/Cosmos-Reason2-2B`. These are valuable negative results because they identify the remaining infrastructure needed before comparing GR00T against pi0.5.

Changes to hypothesis/objective. The proposal originally targeted broad VLA quantization across both LLM and diffusion components. The revised goal is more precise: characterize where static KV quantization fails in flow-matching VLAs, then test a lightweight mitigation such as per-step scaling, rotated KV quantization, or separate prefix/action calibration. pi0.5 is the primary model because its prefix KV cache gives a clean TurboQuant-style stationary component, while its action-expert denoising stream gives a contrasting drifting component. GR00T remains secondary and is used to test whether cross-attention DiT blocks show stronger timestep-dependent drift.

Concrete next steps. First, finish gated Hugging Face access and run one GR00T LIBERO-Spatial smoke evaluation. Second, add KV loggers at the pi0.5 Gemma attention projections and GR00T DiT cross-attention projections. Third, collect one or two LIBERO-Spatial tasks with FP16 tensors, static fake-quant tensors, and per-step/rotated quant tensors. Fourth, report peak VRAM using `nvidia-smi`, task success, static quantization MSE/SQNR, and drift metrics such as RMS drift and Jensen-Shannon histogram distance across denoising steps. The final comparison table will be FP16 baseline versus static quantization versus the proposed mitigation on LIBERO-Spatial.

Figure plan. The milestone figure should show that later denoising steps have a different KV distribution than the calibration step, and that a static quantizer calibrated at step 0 produces larger error than a per-step or rotated quantizer. This directly operationalizes the “problematic drift” claim from the proposal and recent VLA quantization papers.